

Question				Marking details		Marks Available					
						AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Alpha radiation → electromagnetic wave Beta radiation → helium nuclei Gamma radiation → high energy electrons	3 correct 2 marks 1 or 2 correct 1 mark 0 correct 0 mark	2			2		
	(b)			Indicative content: Collecting data Measure the background count of the room for 2 minutes. Using the tongs hold a source close, e.g. 2-3 cm to the GM tube and record the count for the same time. Now place the piece of paper between the source and GM tube or increase the distance in suitable intervals and record the count for 2 minutes. Repeat this technique for the other 2 sources. Analysing data The alpha source will have a count rate that reduces to background count (or 0 if background subtracted) when paper absorber present or distance increased e.g. 30 cm. The beta source will have a count rate that doesn't change when paper is used as absorber or distance increased e.g. 30 cm. The alpha/beta source will have a count rate that decreased when the paper absorber is inserted or distance increased but is higher than the background count.		3	3		6		6

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					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks Detailed description of both collecting data and analysing data for identification of the 3 sources. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Detailed description of either collecting data or analysing data for identification of the 3 sources. Or limited description of both collecting data and analysing data for identification of 2 of the sources. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Limited description of either collecting data or analysing data for identification of the 3 sources. Or limited description of both collecting data and analysing data for identification of 1 of the sources. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 2 total	5	3	0	8	0	6